

# Companion to Bird & de Moor, Algebra of Programming

## §5. Datatypes in Allegories

### §5.1. Relators

(5.1a)

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from  $R$  to  $S$  IS  $R \sqsubseteq S$ . A **relator**  $F : \mathcal{C} \rightarrow \mathcal{D}$  is a 2-functor between allegories:

$$F(1) = 1 \quad F(RS) = F(R)F(S) \quad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving  $\circ$  is **not** asked for —  $\circ$  is an identity-on-objects involution  $\mathcal{C}^{\circ} \rightarrow \mathcal{C}$ , no part of the 2-category.

(5.1b)

the statement

- 1  $\text{Map}(f) \implies \text{Map}(F(f))$  and  $\text{Map}(f) \implies F(f^\circ) = F(f)^\circ$ . Lemma 5.1
- 2 Over a **tabular** allegory a functor is a relator  $\iff$  it preserves  $\circ$ . Theorem 5.1 the book's reverse direction (preserving  $\circ \implies$  relator) needs the source to be **Rel**: the counterexample over a tabular allegory that is not **Rel** shows it can fail
- 3  $F(R^\circ) = F(R)^\circ$  for every  $R$ , so  $F(R)^\circ$  needs no bracket.
- 4 Two relators agreeing on maps are equal. Corollary 5.1
- 5  $\text{Coreflexive}(X) \wedge \text{Coreflexive}(Y) \implies F(X \cap Y) = F(X) \cap F(Y)$  Ex 5.2
- 6  $F(R \cap S) \sqsubseteq F(R) \cap F(S)$ , and strictly. Ex 5.2
- 7  $\text{PreservesRecip}(F) \implies F(\text{dom}(R)) = \text{dom}(F(R))$  Ex 5.5

The **power relator**  $P \multimap xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$  — is where the fourth is strict: for  $R = \{(a_1, b_1), (a_2, b_2)\}$  and  $S = \{(a_1, b_2), (a_2, b_1)\}$  the pair  $(\{a_1, a_2\}, \{b_1, b_2\})$  is in  $P(R) \cap P(S)$ , while  $R \cap S = \emptyset$ .

### §5.2. Relational products

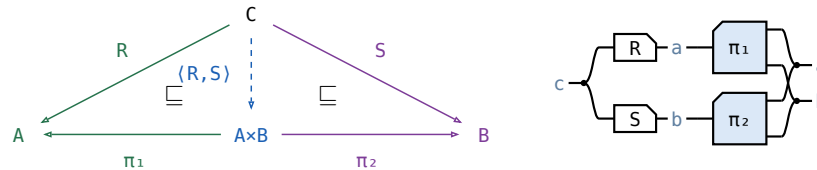
(5.1)

The **fork** of  $R : \mathcal{C} \rightarrow \mathcal{A}$  and  $S : \mathcal{C} \rightarrow \mathcal{B}$  is  $\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ$ , where  $(\pi_1, \pi_2)$  is the tabulation of  $\tau$ .

(5.2a)

$$\langle R, S \rangle \pi_1 = \text{dom}(S) \ R \quad \langle R, S \rangle \pi_2 = \text{dom}(R) \ S \quad (5.6) \quad (5.7)$$

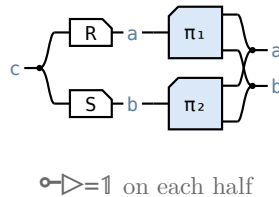
(5.2b)



A domain is coreflexive, so  $\langle R, S \rangle \pi_1 \sqsubseteq R$ , with equality exactly when  $S$  is entire; for maps both triangles commute and  $\langle f, g \rangle$  is unique. In **Rel**,  $c \in \langle R, S \rangle (a, b)$  iff  $c \ R \ a$  and  $c \ S \ b$  — copy  $c$ , then  $R$  on one strand and  $S$  on the other, which is  $\triangleleft (R \otimes S)$  on the right.

No  $\circ$  survives the translation.  $\pi_1 = 1 \otimes \bullet$  discards the second component, so  $\pi_1^\circ = 1 \otimes \bullet$  **creates** one out of nothing, and  $\cap$  is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:

(5.2c)



#### §5.2.1. Relational product $R \times S$

(5.2)

$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$ , a relator in each argument but no longer a categorical product.

(5.2.1a)



Right-then-up is  $(R \times S)\pi_1$ , up-then-right is  $\pi_1 R$ , and  $(R \times S)\pi_1 \sqsubseteq \pi_1 R$ , equality when  $S$  is entire. In  $\mathbf{Rel}$ ,  $(c, d) \in (R \times S) \iff (a, b) \in R$  and  $(c, d) \in S$  — two strands side by side, no copy dot:  $R \times S = R \otimes S$ .

### §5.2.2. Absorption

For  $X : E \rightarrow C$  and  $Y : E \rightarrow D$ ,  $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$ . Both sides are this picture:



(5.3)

(5.2.2a)

|   | the law                                                                                                                                                                                         | picture |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1 | $\langle X, Y \rangle (R, S)^\circ = (X R^\circ) \cap (Y S^\circ)$                                                                                                                              |         |
| 2 | $\langle R, S \rangle^\circ \langle U, V \rangle \sqsubseteq (R^\circ U) \times (S^\circ V)$                                                                                                    |         |
| 3 | $\text{Map}(f) \implies f \langle R, S \rangle = \langle f R, f S \rangle$<br>$f$ a map; it fails for an arbitrary arrow;                                                                       |         |
| 4 | $X \langle R, S \rangle \sqsubseteq \langle X R, X S \rangle$<br>the half of the row above that holds for every $X$                                                                             |         |
| 5 | $F(R \times S) \text{ unzip}(F) \sqsubseteq \text{unzip}(F) (F(R) \times F(S))$<br>$\text{unzip}(F) \triangleq \langle F(\pi_1), F(\pi_2) \rangle$ ; only $\sqsubseteq$ for an arbitrary $R, S$ |         |
| 6 | $g = \text{curry}(f) \iff (g \times 1) \text{ eval} = f$<br>reading $\times$ as the relational product, does $\mathbf{Rel}$ have exponentials?                                                  |         |

## §5.3. Relational coproducts

(5.3a)

|   | the statement                                                                                                                                                                                              | picture |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|   | $R : a_1 \rightarrow c, S : a_2 \rightarrow c$                                                                                                                                                             |         |
| 1 | $[R, S] \triangleq (l^\circ R) \cup (r^\circ S)$<br>(5.9) The tape is the union — a particle entering at $A+B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint. |         |
| 2 | $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$                                                                                                                                                  |         |
| 3 | $l [R, S] = R, r [R, S] = S$<br>$l T = R \wedge r T = S \implies T = [R, S]$                                                                                                                               |         |
| 4 | $U : a_1 \rightarrow d, V : a_2 \rightarrow d$<br>$[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S)$ (5.11)                                                                                              |         |
| 5 | $Z : c \rightarrow d$<br>$[R, S] Z = [R Z, S Z]$                                                                                                                                                           |         |

(5.3b)

|   | the statement                                      | picture                                                                                                                                                           |
|---|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | $R : a_1 \rightarrow b_1, S : a_2 \rightarrow b_2$ |                                                                                                                                                                   |
| 1 | $R + S \triangleq [R l, S r]$ (5.10)               |                                                                                                                                                                   |
| 2 | $l l^\circ = 1, r r^\circ = 1$ Ex 5.12             | $a_1 \text{---} l \text{---} a \text{---} l \text{---} a_1 = a_1 \text{---} a_1$ $a_2 \text{---} r \text{---} a \text{---} r \text{---} a_2 = a_2 \text{---} a_2$ |
| 3 | $l r^\circ = 0, r l^\circ = 0$ Ex 5.12             | $a_1 \text{---} l \text{---} a \text{---} r \text{---} a_2 = a_1 \text{---} 0 \text{---} a_2$                                                                     |
| 4 | $(l^\circ l) \cup (r^\circ r) = 1$ Ex 5.12         |                                                                                                                                                                   |

### §5.3.1. $[R, S] \triangleq [\frac{R}{\exists}, \frac{S}{\exists}] \exists$

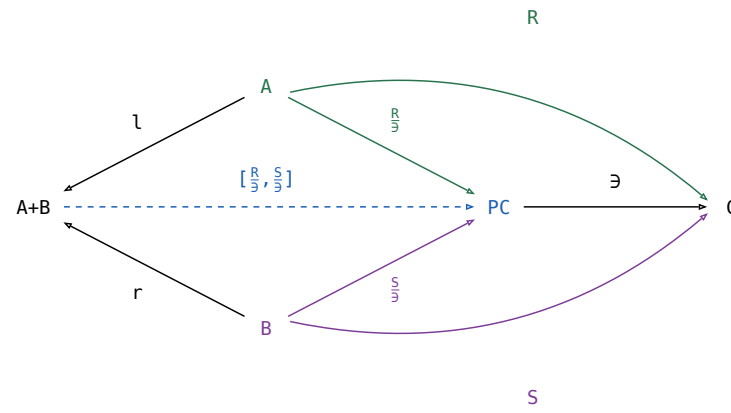
The universal-property row is not free:  $l, r$  were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because  $\exists$  sends an arrow  $A \rightarrow C$  to a map  $A \rightarrow PC$  reversibly, so the map coproduct can be applied underneath it. For any  $T : A+B \rightarrow C$ ,

(5.3.1a)



(5.3a)

(5.3.1b)



Nothing here holds only up to  $\cong$ : every triangle commutes on the nose, which is the difference from the fork above. The border spells  $[R, S] = [\frac{R}{S}, \frac{S}{S}] \exists$ , and pushing  $\exists$  into the union that  $[\cdot, \cdot]$  on maps already is turns that back into the definition,  $(\bar{l} \circ \frac{R}{S} \cup \bar{r} \circ \frac{S}{S}) \exists = \bar{l} \circ R \cup \bar{r} \circ S$ .

(5.3.1c)

| the law                                                                                                                                                                                                                     | picture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 prove (5.11), and say why duality does not carry it over from the product law                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2 $(R + S) \cap ([P, Q] [U, V]^\circ) = (R \cap (P \cup^\circ)) + (S \cap (Q \cup^\circ))$<br>$[P, Q] [U, V]^\circ$ is a full $2 \times 2$ of composites; $R+S$ is diagonal, so the meet cuts the two off-diagonal branches | <p>The diagram shows a complex string diagram representing the law. It features a large bracket on the left labeled <math>s</math> and a large bracket on the right labeled <math>b_1 = s</math>. The diagram is composed of several sub-diagrams enclosed in red boxes, connected by lines. The sub-diagrams include nodes labeled <math>l, r, P, Q, U, V, R, S</math> and arrows labeled <math>a_1, a_2, b_1, b_2, c</math>. The diagram illustrates the relationship between the coproduct <math>R + S</math> and the meet <math>\cap</math> of the composite <math>[P, Q] [U, V]^\circ</math>.</p> |

§5.4. The power relator  $P(R)$ 

(5.4a)

For  $R : A \rightarrow B$ ,

$$P(R) \triangleq (\epsilon \setminus (R \epsilon)) \cap ((\exists R) / \exists)$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R) / \exists) \cap (\exists / (\exists R))^\circ$$

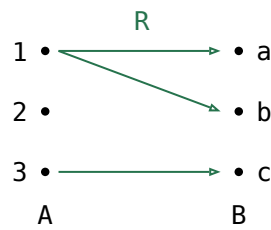
$$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$$

Every element of  $xs$  is related by  $R$  to some element of  $ys$ , and conversely.

(5.4b)

|                                         | in Rel                                                                                              | in words                                                                       | $\text{im}(xs)$              | partner                                       |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------|-----------------------------------------------|
| 1 $(\exists R) / \exists$               | $\forall y \in ys. \exists x \in xs. x \ R \ y$                                                     | $\forall y. \text{ some } xs \ R \ y$                                          | $ys \subseteq \text{im}(xs)$ | every $y$ has a partner                       |
| 2 $(\exists / (\exists R))^\circ$       | $\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$                                     | every $y$ with some $xs \ R \ y$ is in $ys$                                    | $ys \supseteq \text{im}(xs)$ | $ys$ leaves out no partner                    |
| 3 $E(R)$                                | $ys = \{y \mid \exists x \in xs. x \ R \ y\}$                                                       | $ys = \text{every } y \text{ with some } xs \ R \ y$                           | $ys = \text{im}(xs)$         | every $y$ has a partner, and none is left out |
| 4 $((\exists R^\circ) / \exists)^\circ$ | $\forall x \in xs. \exists y \in ys. x \ R \ y$                                                     | $\forall x. x \ R \ \text{some } ys$                                           | —                            | every $x$ has a partner                       |
| 5 $P(R)$                                | $\forall y \in ys. \exists x \in xs. x \ R \ y$ and $\forall x \in xs. \exists y \in ys. x \ R \ y$ | $\forall y. \text{ some } xs \ R \ y$ and $\forall x. x \ R \ \text{some } ys$ | —                            | every $x$ and every $y$ has a partner         |

(5.4c)



| $xs$            | $(\exists R) / \exists$ | $(\exists / (\exists R))^\circ$ | $E(R)$      | $((\exists R^\circ) / \exists)^\circ$ | $P(R)$        |
|-----------------|-------------------------|---------------------------------|-------------|---------------------------------------|---------------|
| 1 $\emptyset$   | $\emptyset$             | all 8 subsets of $abc$          | $\emptyset$ | all 8 subsets of $abc$                | $\emptyset$   |
| 2 $\{1\}$       | $\emptyset, a, b, ab$   | $ab, abc$                       | $ab$        | $a, b, ab, ac, bc, abc$               | $a, b, ab$    |
| 3 $\{2\}$       | $\emptyset$             | all 8 subsets of $abc$          | $\emptyset$ | none                                  | none          |
| 4 $\{3\}$       | $\emptyset, c$          | $c, ac, bc, abc$                | $c$         | $c, ac, bc, abc$                      | $c$           |
| 5 $\{1, 2\}$    | $\emptyset, a, b, ab$   | $ab, abc$                       | $ab$        | none                                  | none          |
| 6 $\{1, 3\}$    | all 8 subsets of $abc$  | $abc$                           | $abc$       | $ac, bc, abc$                         | $ac, bc, abc$ |
| 7 $\{2, 3\}$    | $\emptyset, c$          | $c, ac, bc, abc$                | $c$         | none                                  | none          |
| 8 $\{1, 2, 3\}$ | all 8 subsets of $abc$  | $abc$                           | $abc$       | none                                  | none          |

(5.4d)

| law                                                                                                                  | the reading                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $X \sqsubseteq P(R) \iff X \sqsubseteq \exists \exists R \text{ and } X^\circ \sqsubseteq \exists \exists R^\circ$ | One containment, and the same one at $R^\circ$ — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$ , and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$ .                                                                                                                  |
| 2 $P(1) = 1$                                                                                                         | The straightness axiom verbatim: extensionality is $P$ 's unit law.                                                                                                                                                                                                                                            |
| 3 $\text{Map}(f) \implies P(f) = E(f)$                                                                               | In Rel, $xs \ P(f) \ ys \iff ys = \{f(a) \mid a \in xs\}$ . The half at $f^\circ$ says every $a \in xs$ has its $f(a)$ on $ys$ ; $f$ has just the one image per $a$ , so that already says $ys$ contains everything $xs$ reaches, which is the fraction's second half. For a map the two definitions coincide. |
| 4 $P(R \ S) = P(R) \ P(S)$                                                                                           | $\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$ .                                                                                                                                                    |

§5.4.1. Type relator

(5.4.1a)

Let  $F$  be a binary relator with initial type  $(\alpha, T)$ , so  $T$  is a type functor.  $F(R, S)$  is its action on a pair, and  $F(X)$  abbreviates  $F(1, X)$ , the  $F$  of the reduce section. For every object  $A$  the initial algebra is  $\alpha : F(A, TA) \rightarrow TA$ , among the maps.  $T$  acts on an arrow  $R : A \rightarrow B$  by

$$T(R) = \langle F(R, 1) \ \alpha \rangle$$

(5.4.1b)

|   |  | name                  | law                                       | what it says                                                                                         |
|---|--|-----------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------|
| 1 |  | the defining equation | $T(R) = \langle F(R, 1) \ \alpha \rangle$ | Rebuild the structure with $\alpha$ , applying $R$ to the parameter on the way.                      |
| 2 |  | type relator          | $T(R)^\circ = T(R^\circ)$                 | A datatype acts on relations, not only on maps — the map of the converse is the converse of the map. |

## §5.5. Relational catamorphisms

(5.5a)

let  $F$  be a relator and has **initial algebra**  $\alpha : F(T) \rightarrow T$  in the subcategory of functions.  $\alpha$  is also initial in the allegory:

### §5.5.1. The defining equation

(5.5.1a)



An  $F$ -algebra is a **weakened** natural transformation. A transformation  $F \Rightarrow 1$  on  $\mathcal{A}$  would need a component  $FX \rightarrow X$  at every object and a commuting square at every arrow, but  $F$ -Algebra only need it works on  $T$  and  $A$ .

(5.5.1b)

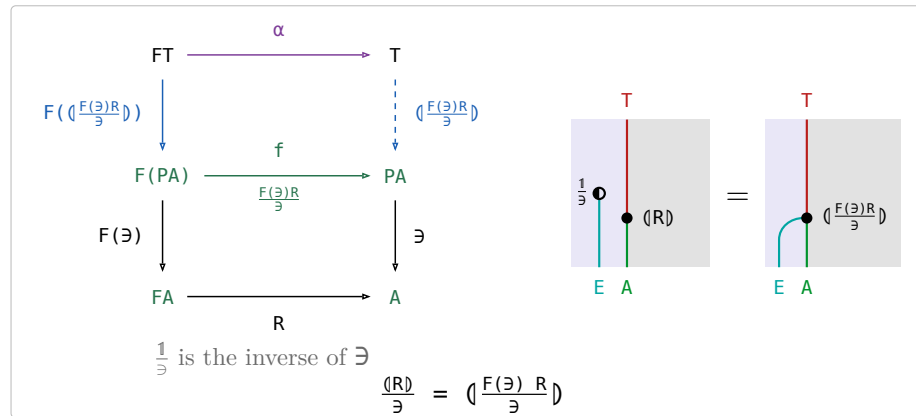
$$([c, f]) = \text{reduce}(c, f)$$

catamorphism: arrows of the form  $([f])$  are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing

|   |                          |                                                                                    |                                                                               |
|---|--------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 1 | <b>part</b>              | <b>Nat</b>                                                                         | <b>[A]</b>                                                                    |
| 2 | datatype                 | $\text{Nat} ::= \text{zero} \mid \text{succ Nat}$                                  | $[A] ::= \text{nil} \mid \text{cons}(A, [A])$                                 |
| 3 | base functor $F$         | $F(X) = 1 + X$                                                                     | $F(X) = 1 + A \times X$                                                       |
| 4 | initial algebra $\alpha$ | $\alpha = [\text{zero}, \text{succ}]$<br>$: 1 + \text{Nat} \rightarrow \text{Nat}$ | $\alpha = [\text{nil}, \text{cons}]$<br>$: 1 + A \times [A] \rightarrow [A]$  |
| 5 | the fold, pointwise      | $([c, f])(\text{zero}) = c$<br>$([c, f])(\text{succ}(n)) = f([c, f](n))$           | $([c, f])(\text{nil}) = c$<br>$([c, f])(\text{cons}(a, x)) = f(a, [c, f](x))$ |

### §5.5.2. $(R) = ([\frac{F(\exists)R}{\exists}])\exists$

(5.5.2a)



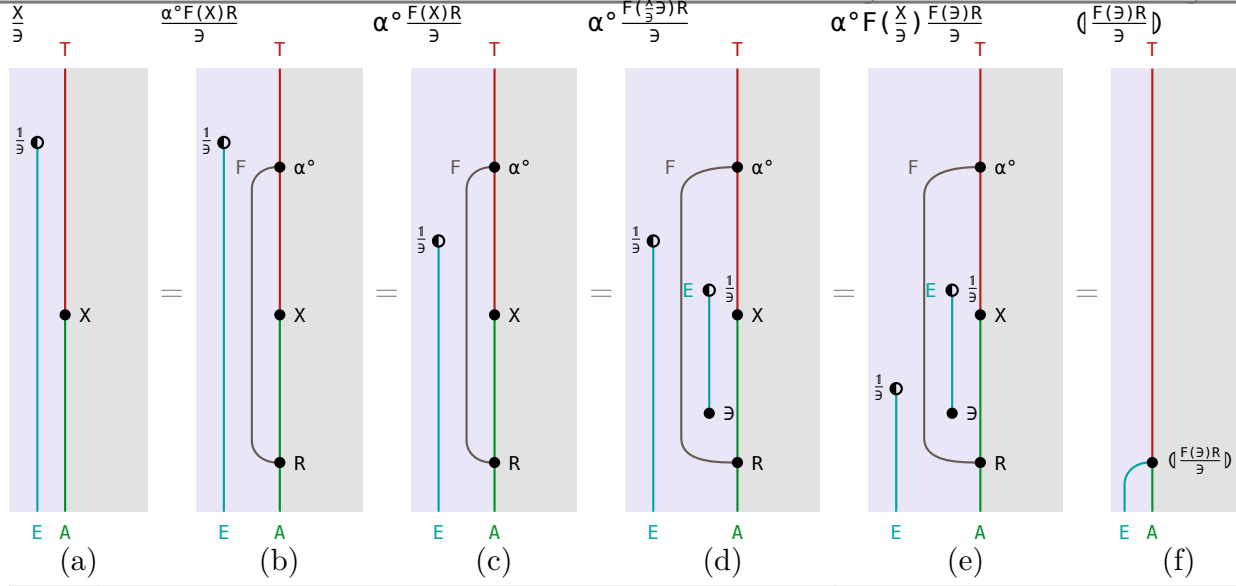
(5.12)

$$\begin{array}{c}
 \boxed{\frac{\alpha X}{F(X)R}} \xleftrightarrow{\cdot \exists \vdash \frac{\exists}{\exists}} \boxed{\frac{\frac{\alpha X}{\exists}}{\frac{F(X)R}{\exists}}} \xleftrightarrow{\cdot \exists \vdash \frac{\exists}{\exists}} \boxed{\frac{\frac{\alpha X}{\exists}}{\frac{F(\frac{X}{\exists})R}{\exists}}} \\
 \xleftrightarrow{\text{relator, fusion twice}} \boxed{\frac{\frac{\alpha}{\frac{X}{\exists}}}{\frac{F(\frac{X}{\exists})}{\frac{F(\exists)R}{\exists}}}} \xleftrightarrow{\text{reduce of maps}} \boxed{\frac{\frac{X}{\exists}}{([\frac{F(\exists)R}{\exists}] \exists)}} \xleftrightarrow{\cdot \exists \vdash \frac{\exists}{\exists}} \boxed{\frac{X}{([\frac{F(\exists)R}{\exists}] \exists)}}
 \end{array}$$

the last two rows at  $X := (R)$ :

$$\alpha X = F(X) R \iff \frac{X}{\exists} = \langle \frac{F(\exists) R}{\exists} \rangle \quad (5.5.2b)$$

in a tabular allegory, with  $F$  a relator and  $\alpha : F(T) \rightarrow T$  its initial algebra: a relation  $X$  out of  $T$  satisfies the fold equation  $\alpha X = F(X) R$  of  $R$  exactly when  $\frac{X}{\exists}$  is the fold of the map  $\frac{F(\exists) R}{\exists}$



|     |                                                                |                                                                                                                                                                                                                                                                               |
|-----|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{X}{\exists}$                                            | $\alpha$ iso; $\alpha X = F(X) R$                                                                                                                                                                                                                                             |
| (b) | $= \frac{\alpha^o F(X) R}{\exists}$                            | $\alpha^o$ is a map because $\alpha$ is an iso, so it leaves $\Lambda$ : $\frac{\alpha^o F(X) R}{\exists} = \alpha^o$<br>$\frac{F(X) R}{\exists} \text{ --- Map}(f)$<br>$\implies \frac{f R}{\exists} = f \frac{R}{\exists}$                                                  |
| (c) | $= \alpha^o \frac{F(X) R}{\exists}$                            | $X = \frac{X}{\exists} \text{ --- } \frac{R}{\exists} \text{ --- } \exists = R$                                                                                                                                                                                               |
| (d) | $= \alpha^o \frac{F(\frac{X}{\exists}) R}{\exists}$            | $F(\frac{X}{\exists}) = F(\frac{X}{\exists}) F(\exists)$ as $F$ is a functor, and $F(\frac{X}{\exists})$ is a map (a relator sends maps to maps), so it leaves $\Lambda$ --- $\triangleq$ relator,<br>$\text{Map}(f)$<br>$\implies \frac{f R}{\exists} = f \frac{R}{\exists}$ |
| (e) | $= \alpha^o F(\frac{X}{\exists}) \frac{F(\exists) R}{\exists}$ | fold uniqueness, $\frac{F(\exists) R}{\exists}$ being a map --- (2.6a)                                                                                                                                                                                                        |
| (f) | $= \langle \frac{F(\exists) R}{\exists} \rangle$               |                                                                                                                                                                                                                                                                               |

## §5.6. Combinatorial functions

(5.6a)

|    | definition                                                                                                                                                                                                                                                                   | type                                                                                                                            | note                                                                                                                                       |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | $[A] ::= \text{nil} \mid \text{cons}(A, [A])$                                                                                                                                                                                                                                | $\mathcal{A} \rightarrow \mathcal{A}$                                                                                           | The list type, under the short name it keeps.                                                                                              |
| 2  | $\text{list}(R) = ([\text{nil}, (R \times 1) \text{ cons}])$                                                                                                                                                                                                                 | $[A] \rightarrow [B]$                                                                                                           | The relator's action on $R : A \rightarrow B$ : one $R$ per element, the shape untouched.                                                  |
| 3  | $\text{subseq} = ([\text{nil}, \text{cons} \cup \pi_2])$                                                                                                                                                                                                                     | $[A] \rightarrow [A]$                                                                                                           | $xs \text{ subseq } ys$ : $ys$ is $xs$ with elements dropped — $\text{cons}$ keeps the head, $\pi_2$ drops it.                             |
| 4  | $\text{prefix} = ([\text{nil}, \text{nil} \cup \text{cons}])$<br>$\text{prefix} = \text{cat}^\circ \pi_1 = \text{init}^*$                                                                                                                                                    | $[A] \rightarrow [A]$                                                                                                           | $ys$ is an initial segment of $xs$ ; the first $\text{nil}$ is where it stops early. $\text{init} \triangleq \text{snoc}^\circ \pi_1$ .    |
| 5  | $\text{suffix} = \text{cat}^\circ \pi_2 = \text{tail}^*$                                                                                                                                                                                                                     | $[A] \rightarrow [A]$                                                                                                           | The dual, $\text{tail} \triangleq \text{cons}^\circ \pi_2$ ; as a reduce it needs $\text{snoc}$ -lists.                                    |
| 6  | $\text{partition} = \text{concat}^\circ$                                                                                                                                                                                                                                     | $[A] \rightarrow [[A]]$                                                                                                         | This $\text{cat}$ is restricted to $[A]^+ \times [A] \rightarrow [A]$ , so $ys$ is a list of non-empty segments of $xs$ .                  |
| 7  | $\text{concat} = ([\text{nil}, \text{cat}])$                                                                                                                                                                                                                                 | $[[A]] \rightarrow [A]$                                                                                                         | Joins the segments back up, which is why its converse splits a list.                                                                       |
| 8  | $\text{inits}$                                                                                                                                                                                                                                                               | $[A] \rightarrow [[A]]$                                                                                                         | Implements $\frac{\text{prefix}}{\exists}$ , listing the prefixes by increasing length.                                                    |
| 9  | $\text{tails}$                                                                                                                                                                                                                                                               | $[A] \rightarrow [[A]]$                                                                                                         | Implements $\frac{\text{suffix}}{\exists}$ by decreasing length — the opposite order.                                                      |
| 10 | $\text{cpp}, \text{cpl}, \text{cpr}$                                                                                                                                                                                                                                         | $[A] \times [B] \rightarrow [A \times B]$<br>$[A] \times B \rightarrow [A \times B]$<br>$A \times [B] \rightarrow [A \times B]$ | Every pair $(a, b)$ with $a$ from the first list and $b$ from the second; $\text{cpl}$ , $\text{cpr}$ keep one side a single element.      |
| 11 | $\text{cpp setify} = (\text{setify} \times \text{setify})$<br>$\frac{\exists \times \exists}{\exists}$<br>$\text{cpl setify} = (\text{setify} \times 1) \frac{\exists \times 1}{\exists}$<br>$\text{cpr setify} = (1 \times \text{setify}) \frac{1 \times \exists}{\exists}$ | $[A] \times [B] \rightarrow P(A \times B)$                                                                                      | Forgetting the order with $\text{setify}$ , $\text{cpp}$ is the transpose of $\exists \times \exists$ .                                    |
| 12 | $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$                                                                                                                                                                                                                         | $F(PA) \rightarrow PFA$                                                                                                         | $\text{cp}(F)$ , the transpose of a relator's action on $\exists$ ; $\text{cpp}$ implements it at $F(A) = A \times A$ .                    |
| 13 | $\text{cp}(\text{list}) = ([\frac{[\text{nil}, (\exists \times \exists) \text{ cons}]}{\exists}])$<br>$\frac{[\text{nil}, (\exists \times \exists) \text{ cons}]}{\exists} = [\text{nil} \frac{1}{\exists}, \frac{\exists \times \exists}{\exists} \text{ E}(\text{cons})]$  | $\text{list}(PA) \rightarrow P(\text{list}(A))$                                                                                 | $\text{cp}(\text{list})$ : (5.12) at $\text{list}(\exists) = ([\text{nil}, (\exists \times 1) \text{ cons}])$ , then the algebra expanded. |
| 14 | $\text{cplist} = ([\text{nil wrap}, \text{cpp list}(\text{cons})])$                                                                                                                                                                                                          | $[[A]] \rightarrow [[A]]$                                                                                                       | $\text{cp}(\text{list})$ with every set a list: $\text{cpp}$ in place of $\frac{\exists \times \exists}{\exists}$ .                        |

**glue concat  $\sqsubseteq (1 \times \text{concat}) \text{ cons}$**

(5.6b)

putting **a** on the front of the first segment and flattening (**glue concat**) gives at most flattening the segments and putting **a** on the front of the result (**(1×concat) cons**)

the starting composite



|     |                                                                                                              |                                                    |
|-----|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| (a) | glue concat                                                                                                  | definition of glue                                 |
| (b) | $\triangleq (1 \times \text{cons}^\circ) \text{ assocl } \text{cons} \times 1 \text{ cons } \text{concat}$   | since $\text{concat} = ([\text{nil}, \text{cat}])$ |
| (c) | $= (1 \times \text{cons}^\circ) \text{ assocl } \text{cons} \times 1 (1 \times \text{concat}) \text{ cat}$   | naturality of $\text{assocl}$                      |
| (d) | $= (1 \times (\text{cons}^\circ (1 \times \text{concat}))) \text{ assocl } \text{cons} \times 1 \text{ cat}$ | since $\text{concat} = ([\text{nil}, \text{cat}])$ |



|     |                                                                                                                            |                                                                                                |
|-----|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| (a) | $\sqsubseteq (1 \times \text{concat}) (1 \times \text{cat}^\circ) \text{ assocl } \text{cons} \times 1 \text{ cat}$        | since $(\text{cons} \times 1) \text{ cat} = \text{assocr } (1 \times \text{cat}) \text{ cons}$ |
| (b) | $= (1 \times \text{concat}) (1 \times \text{cat}^\circ) \text{ assocl } \text{assocr } (1 \times \text{cat}) \text{ cons}$ | since $\text{assocl } \text{assocr} = 1$ and $\text{cat}^\circ \text{ cat} \sqsubseteq 1$      |
| (c) | $\sqsubseteq (1 \times \text{concat}) \text{ cons}$                                                                        |                                                                                                |

**§5.6.1.**  $\frac{\text{subseq}}{\exists} = ([\text{nil } \frac{1}{\exists}, (\frac{1 \times \exists}{\exists} \text{ E}(\text{cons}), \pi_2) \text{ cup}])$

(5.6.1a)

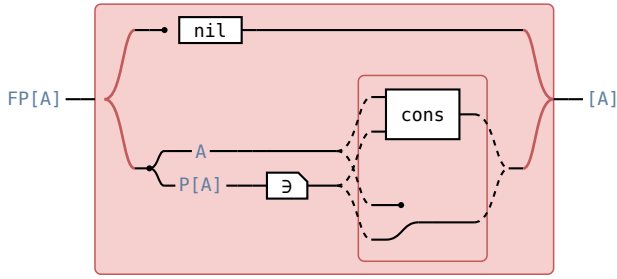
$\text{cup} \triangleq \frac{\pi_1 \exists \cup \pi_2 \exists}{\exists} : \text{PA} \times \text{PA} \rightarrow \text{PA}$ , so  $\frac{R \cup S}{\exists} = (\frac{R}{\exists}, \frac{S}{\exists}) \text{ cup}$ .

$$\frac{F(\exists) [\text{nil}, \text{cons } u \pi_2]}{\exists} = [\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists) (\text{cons } u \pi_2)}{\exists}] \quad (5.6.1b)$$

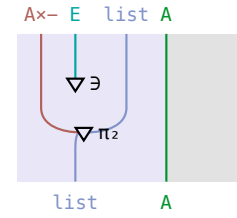
the set of lists the algebra builds is, from nothing, just **nil**, and from a head and a set of tails, every tail in the set with the head put on or left off — (5.12) at **subseq**=( $[\text{nil}, \text{cons } u \pi_2]$ ), **subseq** = ( $[\text{nil}, \text{cons } u \pi_2]$ ) .

**circuit** — the fork is  $F([A])=1+A \times [A]$ : **nil** above, the pair below

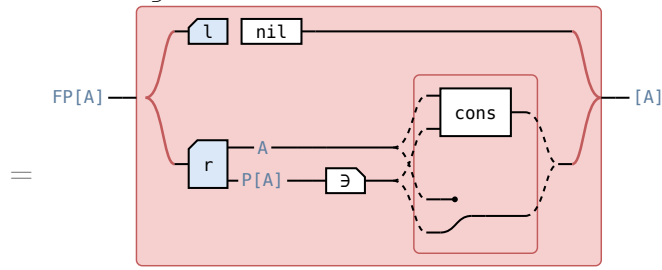
**Hinze–Marsden**



$$\frac{F(\exists) [\text{nil}, \text{cons } u \pi_2]}{\exists}$$

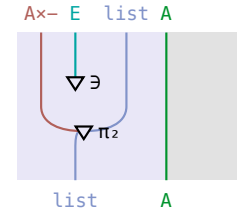


the  $\pi_2$  operand of  $\text{cons } u \pi_2$  under the  $1 \times \exists$  summand of  $F(\exists)$ , i.e.  $(1 \times \exists) \pi_2$

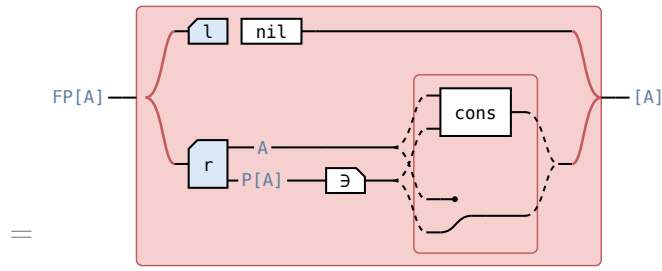


$$\frac{(1+1 \times \exists) [\text{nil}, \text{cons } u \pi_2]}{\exists}$$

$$F(X)=1+A \times X \quad (5.6a.1)$$

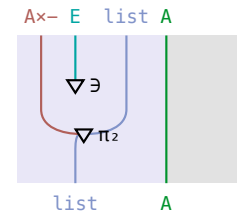


the same operand under  $1+1 \times \exists$ , whose  $1 \times \exists$  summand it sits in

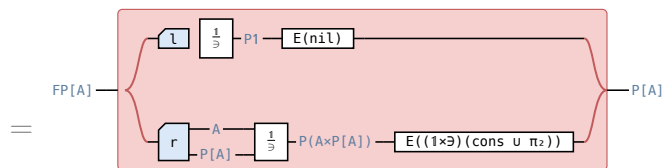


$$\frac{[\text{nil}, (1 \times \exists) (\text{cons } u \pi_2)]}{\exists}$$

$$R+S \triangleq [Rl, Sr], l[R, S]=R, r[R, S]=S \quad [R, S] \triangleq (l \circ R) \cup (r \circ S), l[R, S]=R, r[R, S]=S, lT=R \wedge rT=S \Rightarrow T=[R, S]$$

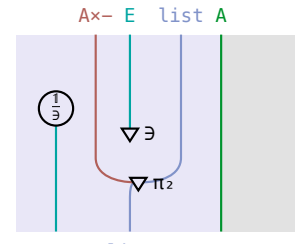


the  $\pi_2$  operand of the second arm  $(1 \times \exists) (\text{cons } u \pi_2)$

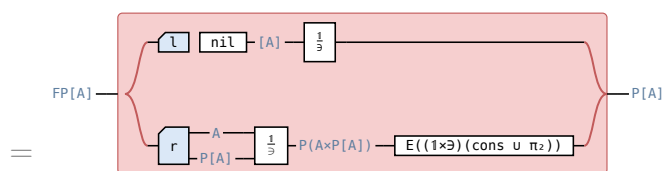


$$\frac{[\text{nil}, (1 \times \exists) (\text{cons } u \pi_2)]}{\exists}$$

$$(5.3.1a) \text{ at } T := [\text{nil}, (1 \times \exists) (\text{cons } u \pi_2)]$$



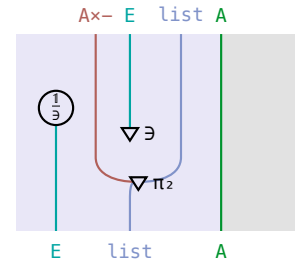
the  $\pi_2$  operand under its  $1 \times \exists$ , the arm (5.6.1c)'s square rewrites,  $(1 \times \exists) \pi_2 = \pi_2 \exists$



$$\frac{[\text{nil } \frac{1}{\exists}, (1 \times \exists) (\text{cons } u \pi_2)]}{\exists}$$

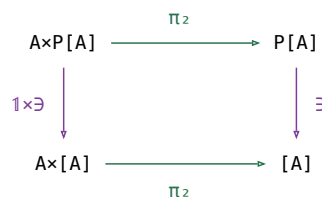
$$\text{Map}(f) \wedge 1 \square = \exists \square$$

$$\Rightarrow \frac{f}{\exists} = f \frac{1}{\exists}, \frac{f}{\exists} = f \frac{1}{\exists} \text{ for } f \text{ a map, at } f := \text{nil}$$

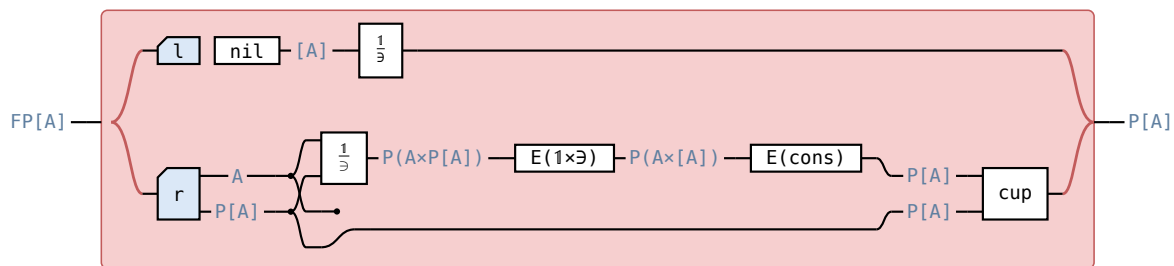


the same operand; the two rows differ only in the **nil** arm

(5.6.1c)



| $\frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists} = \left( \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup}$ <p>power transpose of join: the power transpose of the join of two relations is <math>\left( \frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup}</math>, where <b>cup</b> is the function that returns the union of two sets</p>        | (5.6.1d)                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <p><b>circuit</b></p> <p><math display="block">\frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists}</math></p> <p>(5.6.1b) <math>\frac{F(\exists) [\text{nil}, \text{cons} \cup \pi_2]}{\exists} = [\text{nil} \frac{1}{\exists}, \frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists}]</math> 's second branch</p>                                                            | <p><b>Hinze–Marsden</b></p> <p>the <b>cons</b> operand of <b>cons</b> <math>\cup</math> <math>\pi_2</math></p>     |
| <p><math display="block">\frac{(1 \times \exists) \text{cons} \cup (1 \times \exists) \pi_2}{\exists}</math></p> <p><math>T(X_1 \cup X_2) = TX_1 \cup TX_2</math> — (0.2a.8)</p>                                                                                                                                                                                                             | <p>the <math>\pi_2</math> operand of <math>(1 \times \exists) \text{cons} \cup (1 \times \exists) \pi_2</math></p> |
| <p><math display="block">\frac{(1 \times \exists) \text{cons} \cup \pi_2 \exists}{\exists}</math></p> <p><math>(1 \times \exists) \pi_2 = \pi_2 \exists</math> — (5.2.1a) at <math>\pi_2</math>, an equality because <math>1</math> is entire</p>                                                                                                                                            | <p>the <math>\pi_2</math> operand of <math>(1 \times \exists) \text{cons} \cup \pi_2 \exists</math></p>            |
| <p><math display="block">\frac{(1 \times \exists) \text{cons} \cup \pi_2 \exists}{\exists} \text{cup}</math></p> <p><math>\frac{R \cup S}{\exists} = \left( \frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup} \triangleq \text{cup}</math></p>                                                                                                                                         | <p>the <b>cons</b> operand under its <math>1\% \exists</math></p>                                                  |
| <p><math display="block">\frac{(1 \times \exists) E(\text{cons}), \pi_2}{\exists} \text{cup}</math></p> <p><math>\frac{S}{\exists} E(R) = \frac{S}{\exists} R</math>, absorption <math>\frac{S}{\exists} E(R) = \frac{SR}{\exists}</math> at <math>S := 1 \times \exists, R := \text{cons}</math>; fusion and <math>\frac{\exists}{\exists} = 1</math> on the <math>\pi_2</math> operand</p> | <p>the <math>\pi_2</math> operand, bare <math>\pi_2</math></p>                                                     |



$$[\text{nil} \frac{1}{\exists}, \frac{(1 \times \exists) E(\text{cons}), \pi_2}{\exists} \text{cup}]$$

which writes **Pcons**; **cons** is a map, and there  $P(\text{cons}) = E(\text{cons})$  —  $\text{Map}(f)$

$$\Rightarrow P(f) = E(f) .$$

(5.6.1e)

## §5.7. Lax natural transformations (LaT)

(5.7a)

For relators  $G, F : \mathcal{C} \rightarrow \mathcal{D}$  and components  $\varphi_A : GA \rightarrow FA$ ,  $\varphi$  is **lax at**  $R : A \rightarrow B$  when  $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$  — both sides  $GA \rightarrow FB$ , the left through the component at  $B$ , the right through the one at  $A$ .

$\varphi : G \Rightarrow F$  is a **lax natural transformation** (LaT) when it is lax at **every**  $R$ . (5.13)

Lax at every **map** already gives LaT, and at a map the inequation is an equality  $G(f)\varphi = \varphi F(f)$ : laxness is about relations only.

(5.7b)



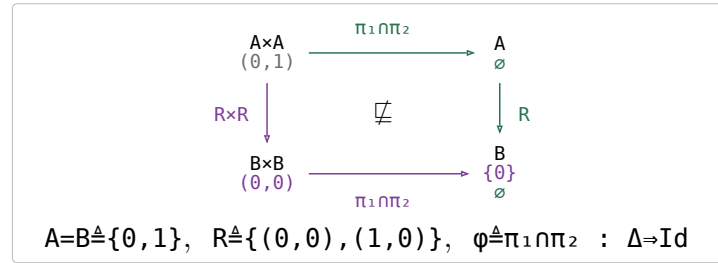
(5.7c)

|                                                                                                                                                                                                                                                                                                                                                                      | closed under | commutative diagram                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | string diagram |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 1<br>composition<br>$\psi\phi$                                                                                                                                                                                                                                                                                                                                       |              | <p><math>H(R)\psi_B \sqsubseteq \psi_A G(R)</math> and <math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> give <math>H(R)(\psi_B\phi_B) \sqsubseteq (\psi_A\phi_A)F(R)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |
| 2<br>horizontal composition<br>$\chi\circ\phi$                                                                                                                                                                                                                                                                                                                       |              | <p><math>\phi : G \Rightarrow F</math> with <math>G, F : \mathcal{C} \rightarrow \mathcal{D}</math> and <math>\chi : L \Rightarrow K</math> with <math>L, K : \mathcal{D} \rightarrow \mathcal{E}</math> give <math>\chi \circ \phi : L \circ G \Rightarrow K \circ F</math>, the family <math>A \mapsto \chi_{GA} K(\phi_A)</math><br/> <math>L(G(R))\chi_{GB} \sqsubseteq \chi_{GA} K(G(R))</math> is <math>\chi</math> lax at <math>G(R)</math>; then<br/> <math>K(G(R)\phi_B) \sqsubseteq K(\phi_A F(R))</math> is <math>K</math> applied to <math>\phi</math>'s own inequation;<br/> <math>\chi := \mathbb{1}_K</math> gives <math>K(\phi) : K \circ G \Rightarrow K \circ F</math>, <math>K</math> composed on the outside; <math>\phi := \mathbb{1}_G</math> gives <math>\chi G : L \circ G \Rightarrow K \circ G</math>, <math>G</math> composed on the inside<br/> two candidates, <math>\chi_{GA} K(\phi_A)</math> and <math>L(\phi_A)\chi_{FA}</math>, ordered by <math>\sqsubseteq</math>; this row is the first</p> |                |
| 3<br>union<br>$\phi \cup \psi$                                                                                                                                                                                                                                                                                                                                       |              | <p><math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> and <math>G(R)\psi_B \sqsubseteq \psi_A F(R)</math> give <math>G(R)(\phi_B \cup \psi_B) \sqsubseteq (\phi_A \cup \psi_A)F(R)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |
| 4<br>a relator $K$<br>$K(\phi)$                                                                                                                                                                                                                                                                                                                                      |              | <p><math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> gives <math>K(G(R))K(\phi_B) \sqsubseteq K(\phi_A)K(F(R))</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                |
| 5<br>product<br>$\phi \times \psi$                                                                                                                                                                                                                                                                                                                                   |              | <p><math>\phi : G \Rightarrow F</math> and <math>\psi : G' \Rightarrow F'</math> give <math>\phi \times \psi : G \times G' \Rightarrow F \times F'</math><br/> <math>(R \times S)(U \times V) = (RU) \times (SV)</math> and monotonicity in both slots — the row above's <math>K</math> applied to the inequation, run on a bifunctor; the fork is the derived case <math>\langle \phi, \psi \rangle = \triangleleft(\phi \times \psi)</math>, and its ONE cost is the lax copy law <math>R \triangleleft \triangleleft (R \times R)</math> — (0.4a)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |
| 6<br>coproduct<br>$\phi + \psi$                                                                                                                                                                                                                                                                                                                                      |              | <p><math>\phi : G \Rightarrow F</math> and <math>\psi : G' \Rightarrow F'</math> give <math>\phi + \psi : G + G' \Rightarrow F + F'</math><br/> <math>(R + S)(U + V) = (RU) + (SV)</math> and monotonicity in both slots; the co-fork is the derived case <math>[\phi, \psi] = (\phi + \psi) \triangleright</math>, and <math>\triangleright</math> costs nothing<br/> the step it would need is <math>(\phi_A F(R)) \sqcap (\psi_A F(R)) \sqsubseteq (\phi \sqcap \psi)_A F(R)</math>, the wrong direction of (0.6a)<br/> a counterexample: (5.7.1a)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                |
| 7<br>meet — <b>fails</b><br>$\phi \sqcap \psi$                                                                                                                                                                                                                                                                                                                       |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |
| 8 Each row is the general slide $R_A X \sqsubseteq X' R_B$ and $R_B Y \sqsubseteq Y' R_C$ give $R_A(XY) \sqsubseteq (X'Y')R_C$ at $R_A, R_B := G(R), F(R)$ and $X, X' := \phi_B, \phi_A$ , quantified over $R$ ; at $X' = X$ with $R_A, R_B$ endorelations it is the monotonicity reading instead — the $\forall R$ sits outside the law, and is the only difference |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |

## §5.7.1. meet is not closed in LaT

counterexample —  $\varphi, \psi : \mathbf{G} \Rightarrow \mathbf{F}$  lax natural does NOT give  $\varphi \cap \psi$  lax natural

(5.7.1a)



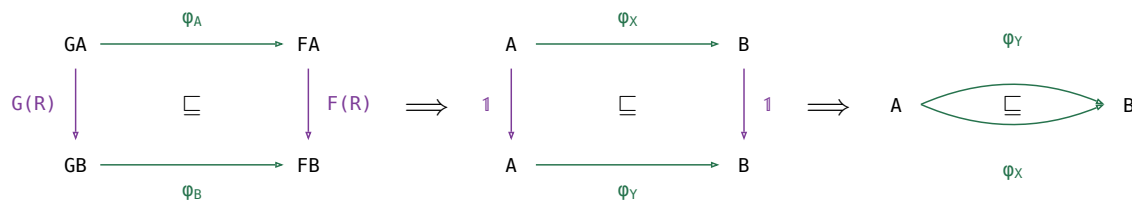
## §5.7.2. Two stacked towers

(5.7.2a)

|                | lower — inside ONE allegory      | upper — between allegories                                                                                                                                                                           |
|----------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 0-cell       | an object $A$                    | an allegory $\mathcal{C}$ — a whole lower tower                                                                                                                                                      |
| 2 1-cell       | a relation $R : A \rightarrow B$ | a relator $F : \mathcal{C} \rightarrow \mathcal{D}$                                                                                                                                                  |
| 3 2-cell       | $R \sqsubseteq S$                | a LaT $\varphi : \mathbf{G} \Rightarrow \mathbf{F} \triangleq \text{lax}$                                                                                                                            |
| 4 order        | $R \sqsubseteq S$                | $\varphi \sqsubseteq \psi$ componentwise                                                                                                                                                             |
| 5 union        | $R \cup S$                       | $\varphi \cup \psi$ , survives $G(R) (\varphi_B \cup \psi_B) \sqsubseteq (\varphi_A \cup \psi_A) F(R)$                                                                                               |
| 6 product      | $R \times S \triangleq \times$   | $\varphi \times \psi$ , survives; a TENSOR, not a categorical product; SYMMETRIC (5.2a)<br>$(G(R) \times G'(R)) (\varphi_B \times \psi_B) \sqsubseteq (\varphi_A \times \psi_A) (F(R) \times F'(R))$ |
| 7 coproduct    | $R + S$                          | $\varphi + \psi$ , survives; a BIPRODUCT $(G(R) + G'(R)) (\varphi_B + \psi_B) \sqsubseteq (\varphi_A + \psi_A) (F(R) + F'(R))$                                                                       |
| 8 meet         | $R \cap S$                       | $\varphi \cap \psi$ componentwise, fails (5.7.1a)                                                                                                                                                    |
| 9 converse     | $R^\circ$                        | $\varphi^\circ$ , fails, oplax: $\varphi_A^\circ G(R) \sqsubseteq F(R) \varphi_B^\circ$                                                                                                              |
| 10 zero object | $z$ with $1 \ z = 0$             | the constant relator at such a $z$ , initial and terminal                                                                                                                                            |

## §5.7.3. RES is a special case of LaT

(5.7.3a)



$G \triangleq \text{const } A, F \triangleq \text{const } B$  — the relators  $X \rightarrow A$  and  $X \rightarrow B$ , each sending every arrow to  $1 \triangleq \text{lax}$

with  $\mathcal{C}$  two objects and one non-identity arrow  $X \rightarrow Y$ , a LaT  $\text{const } A \Rightarrow \text{const } B$  is the pair  $\varphi_X, \varphi_Y : A \rightarrow B$  with  $\varphi_Y \sqsubseteq \varphi_X$  — the datum  $R \sqsubseteq S$

not every LaT is constant:  $\exists : \mathbf{P} \Rightarrow \text{Id}$  (5.4d.1)  $\pi_2 : (A \times -) \Rightarrow \text{Id}$  (5.6.1c)  $\triangleleft : \text{Id} \Rightarrow \Delta$  (0.4a)